

```
#include <iostream>

using namespace std;

// Base Class
class Person {
protected:
    string name;
    int age;

public:
    void setPerson(string n, int a) {
        name = n;
        age = a;
    }

    void displayPerson() {
        cout << "Name: " << name << endl;
        cout << "Age: " << age << endl;
    }
};

// Derived Class 1 (Hierarchical Inheritance)
class Student : public Person {
protected:
    int rollNumber;

public:
```

```

void setStudent(string n, int a, int roll) {
    setPerson(n, a);    // calling base class function
    rollNumber = roll;
}

void displayStudent() {
    displayPerson();
    cout << "Roll Number: " << rollNumber << endl;
}
};

// Derived Class 2 (Hierarchical Inheritance)
class Employee : public Person {
protected:
    int employeeID;

public:
    void setEmployee(string n, int a, int id) {
        setPerson(n, a);
        employeeID = id;
    }

    void displayEmployee() {
        displayPerson();
        cout << "Employee ID: " << employeeID << endl;
    }
};

```

// Most Derived Class (Multilevel Inheritance)

class TeachingAssistant : public Student {

protected:

int employeeID;

public:

void setTA(string n, int a, int roll, int id) {

setStudent(n, a, roll); // multilevel: Student → Person

employeeID = id;

}

void displayTA() {

displayStudent();

cout << "Employee ID: " << employeeID << endl;

}

};

int main() {

// Object of most derived class

TeachingAssistant ta;

// Setting values

ta.setTA("Ali", 22, 101, 5001);

// Displaying complete information

```
    cout << "---- Teaching Assistant Information ----" << endl;
    ta.displayTA();

    return 0;
}
```

OUTPUT:

 C:\Users\AAZ.Computer\Desktop\p3.exe

---- Teaching Assistant Information ----

Name: Ali

Age: 22

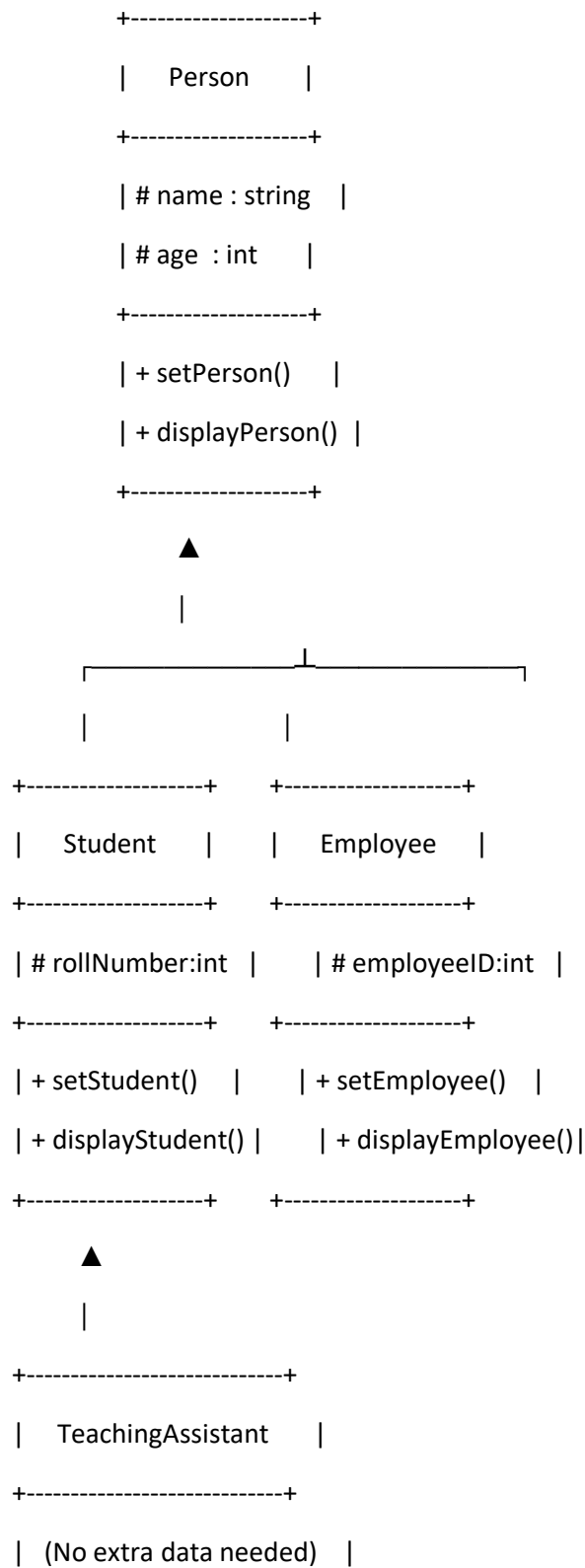
Roll Number: 101

Employee ID: 5001

Process exited after 1.249 seconds with return value 0

Press any key to continue . . .

UML DIAGRAM:



```
+-----+
| + displayTA()      |
+-----+
```

